



FPGA-powered test equipment from Aeroflex

graphics processing units (GPUs).

Vinod Mathews, founder and CEO, Captronic Systems, says that while using FPGA-based equipment, the biggest benefit in speed is especially noticed when you need to do hardware-in-the-loop kind of tests.

Customisability. Initially, the FGAs were very difficult to program due to lack of simplifying tools, and this affected engineers working with COTS FPGA boards a lot. The HDL programming required was very un-intuitive and demanded expertise which most users did not have. Second, there was no hardware platform or architecture, no starting point that the users could leverage to quickly build their systems. The only option had been to start from scratch, from the FPGA chip and figure out the rest of the pieces like IO connectivity, power requirements, interaction with other ASICs and processors (if present), by yourselves. This required a lot of time, effort and money, and was not at all viable, especially with the technology advancing ever so rapidly, and changing the test and measurement requirements with it.

NI LabVIEW is one solution to this problem of customisability. "National Instruments, realising the benefits of a user-programmable FPGA, came up with the LabVIEW FPGA module, which abstracted the complexities of FPGA programming and enabled the user to use the same graphical LabVIEW environment to program the FGAs. Simultaneously, NI also came up with a revolutionary RIO architec-

ture that had an FPGA, a real-time processor and modular IO at its core. It abstracted all hardware design complexities from the user and could be used just as well for test and measurement instrumentation, as for embedded systems," explains Chinmay.

Reduction in both test time and physical footprint. "By embed-

ding FGAs into T&M equipment, test routines can be run at a much faster rate, thereby reducing the testing costs. Since FGAs are highly configurable, newer test cases and functionality can be embedded in a short period of time to allow the equipment manufacturers to reach market on a timely basis," explains Nate Srinath, founder-director, Inxe Technologies.

NI's Chinmay also shares an example of reduction in test time achieved by FPGA-based instruments, in which by synchronising timing of digital control with the on-board FPGA on the NI vector signal transceiver with the RF front end of the instrument, Qualcomm Atheros was able to reduce the test time by 20x over their previous PXI-based solution and reduced the physical footprint for their test system from several different boxes to a single 8-slot PXIe chassis with slots to spare.

Lower time-to-market for equipment vendors. In many parts of the wireless industry, time-to-market is critical. The designer of a real mobile has to worry about many factors such as achieving very low manufacturing costs, long battery life and small size. Cost is of course important for test equipment manufacturers but size and power consumption are less so, and the use of FGAs allows the vendor to deliver functionality in line with when base station manufacturers need it—Aeroflex's first LTE test mobile was delivered in November 2007, some four years before the first networks launched.

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